WHITE BLOOD CELLS (WBCS) OR LEUKOCYTE

They are nucleated cells formed by the bone marrow and lymphoid tissue. The most important function of WBCs is defense mechanism against microorganisms and their toxins.

They are whitish in color. After a short life span, they are destroyed by reticulo-endothelial cells.

WBCS NUMBER VARIES ACCORDING TO:-

1) Species: Man, horse, pig, cattle, sheep and dog → 6.000-10.000/mm^3; camel, cat and goat → 10.000-15.000/mm^3; birds→ 20.000-30.000/mm^3.

2) Sex: Females exhibit higher count.

3) Age: Low in newly born, then gradually increased till maturity.

4) Feeding: After meals, the WBCs number increases (digestive leukocytosis).

WBCS NUMBER VARIES ACCORDING TO:-

5) **Pregnancy:** WBCs number increases in early months of gestation and just before and after parturition.

6) **Exercise:** Muscular exercise increases the WBCs number.

7) **Estrus phase:** During heat, the count of WBCs increases.

8) **Cold weather and cold bath:** Increases WBCs count.

9) **Emotion or increase adrenaline:** Increases WBCs count

Leukocytosis:- "It means increase in WBCs count till 100% of the normal count".

1) **Physiological:-** ↑ in total number of WBCs till 100% of normal counwithout change in the % of each type of the WBCs as in heat, cold weather, emotion and after eating.

2) **Pathological:-** ↑ in total number of WBCs till 100% of normal count with change in the % of certain type of the WBCs at the expense of other types as in systemic infection.

Leukaemia:- "It means increase in WBCs count above 100% of the normal count", as in bone marrow tumor.

Leukopenia:- "It means decrease in WBCs count below 50% of the normal count" as in viral diseases, toxemia , septicemia and typhoid fever.

A) GRANULOCYTES

They have granules in the cytoplasm and have a lobed nucleus. They are formed in the bone marrow under the effect of granulopoietin. According to the staining of the granules, they are divided into 3 types :-

A) Granulocytes

1. Neutrophil (microphage or polymorphnuclear cells)
2. Eosinophils (Acidophils)
3. Basophils (Heparinocyte)

1-NEUTROPHIL

(MICROPHAGE OR POLYMORPHNUCLEAR CELLS):-

Their granules are not stain (neutrophilic) except in birds are acidophilic (stained red) so, it called pseudo-eosinophil or heterophil. The diameter of neutrophil about 7.5-15μ.

Life span: 6 hours.

Nucleus:-

a- Not segmented nucleus (band or stab): Its percentage about 3-5% of the neutrophil, when its % ↑ this called shift to left.

b- Segmented nucleus: The nucleus is divided into several lobes (2-5 lobes) attached by filaments. The number of lobes increases (from 2 to 5) with the age of neutrophil. Segmented neutrophil represented about 95-97% of the neutrophil, when its % ↑ this called shift to right.

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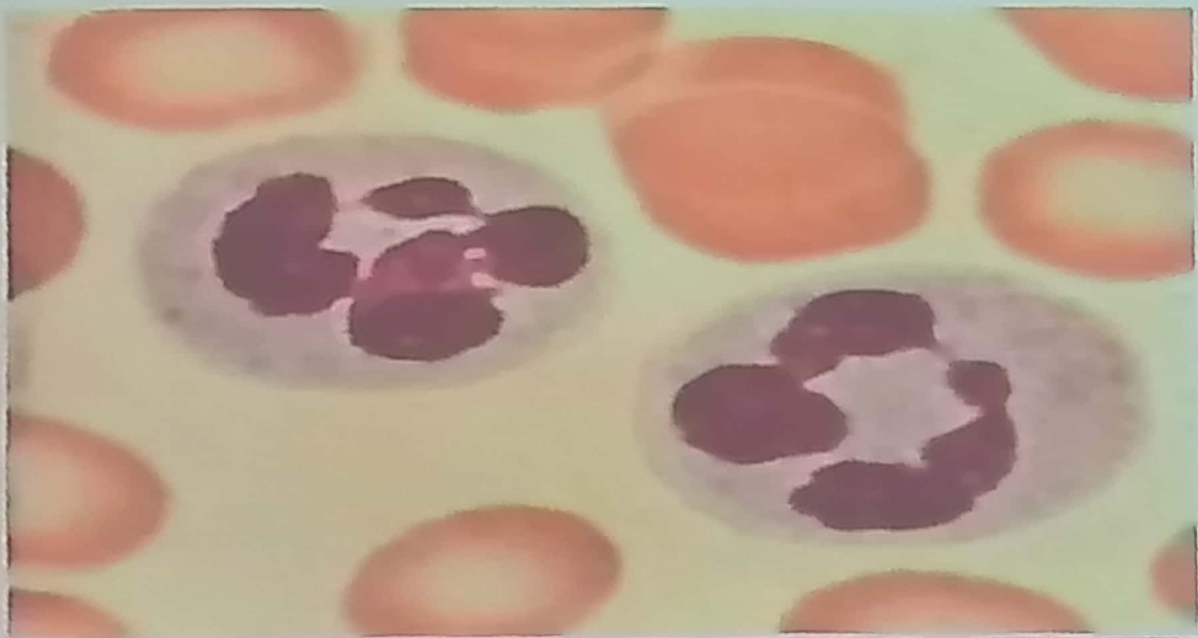
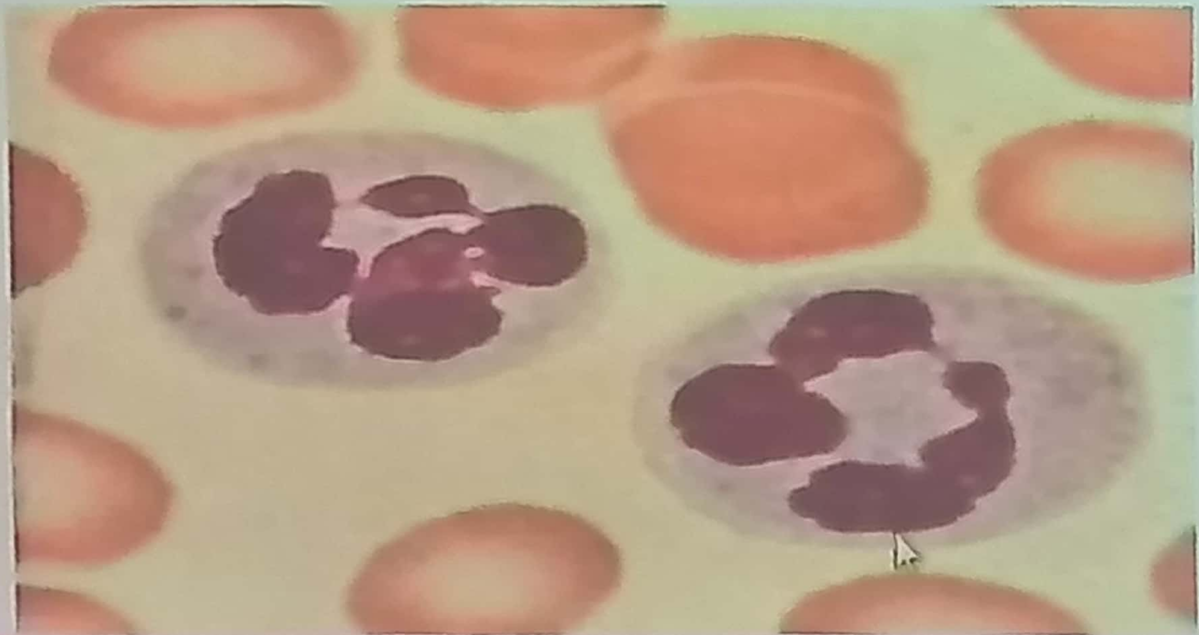
1-NEUTROPHIL
(MICROPHAGE OR POLYMORPHNUCLEAR CELLS):-



1-NEUTROPHIL (MICROPHAGE OR POLYMORPHNUCLEAR CELLS):-

Sex chromatin (Drum stick) is small nuclear expansion seen in some neutrophil of female and they are characteristic of the sex.

The percentage of neutrophil varies according to the species: in man, dog, cat and horse is about 65%, while in cattle, sheep, goat, pig and birds is about 30%.



FUNCTIONS OF NEUTROPHILS:-

It considered the 1st line of defense.

1) **Phagocytosis:-** They are highly phagocytic to small particles e.g. bacteria. Three steps must be done before phagocytosis:-

a) Chemotaxis: The interaction between bacterial products with plasma factors and cells produces a substance called leukotoxine, which make 3 functions:

1- Increase the blood vessels permeability.

2- Attract neutrophils toward the site of inflammation.

3- Stimulate the bone marrow to produce WBCs.

b) Opsonization: A substance called opsonine (type of immunoglobulin, IgG) coat and fix bacteria and to make them tastes for neutrophils to phagocytic it.

FUNCTIONS OF NEUTROPHILS:-

c) Diapedesis: Means the escape of leukocyte from the dilated blood vessel pores. In tissue spaces, neutrophils moves by pseudopodia (like amoeba). The neutrophils engulf microorganisms and digest it by its proteolytic enzymes.

2) **Release leukotriens** which make the same effect of leukotoxine.

3) **Release thromboxane** (a vasoconstrictor and platelet aggregating agent).

4) **Secrete prostaglandins** (which are inflammatory agent).

5) **Secrete defensin** (antibacterial agent).

FUNCTIONS OF NEUTROPHILS:-

 # Killing zone:-

"It is the area around the activated neutrophil" in which all the foreign bodies are destroyed by the following production:-

1- Oxidizing agents as nascent oxygen ($O2-$) & hydrogen peroxide $H2o2$.

2- Oxidizing acids as per-chloric acid (Hclo2) & per-bromic acid (HBro2).

3- Anti-bacterial agents as defensin (which secreted from neutrophil).

FUNCTIONS OF NEUTROPHILS:

 # Toxic neutrophil: Abnormal neutrophils occur in toxic condition, in which it contain blue black granules. In dogs and cat a common indication of toxicity is the presence of vacuoles located in the cytoplasm along the periphery of the neutrophils.

 # Neutrophilia: Means increased number of neutrophils due to systemic & localized acute infections (e.g. appendicitis), lead & mercury poisoning, uremia, tissue destruction, stress and increase ACTH & corticoids.

 # Neutropnia: Means decreased number of neutrophils due to typhoid fever, viral diseases, toxemia and septicemia.

2-EOSINOPHILS

The diameter of eosinophils is about 11-12 μ and their percentage about 2-5%. Its cytoplasm contains pen head red granules (vacuoles like coins in horse). The nucleus is divided into 2-3 lobes.

Functions:(4th line of defense)

1) Anti-allergic & detoxification: They may absorb histamine released in allergic conditions. They

may also detoxify toxins and foreign proteins which increases in these conditions.

2) Release plasminogen which lyses and removal of blood clots by its fibrinolytic effect.

3) They are weak phagocytes.

2-EOSINOPHILS (ACIDOPHILS)

Eosinophilia: Increase number of eosinophils mainly occurs in allergic and parasitic diseases (to destruct the metabolic by-product of the parasites).

Eosinopenia: Decrease number of eosinophils mainly occurs in stress or hyper-activity of adrenal gland or increased level of ACTH or corticoid.

3- BASOPHILS (HEPARINOCYTE)

The diameter of basophils is about 10-15 μ and their percentage about 0.5-1 %. It has a basophilic granules (blue in color), spherical or S-shape nucleus.

↑ with chronic inflammation, liver cirrhosis and some skin diseases.

Functions: (5th line of defense)

1) Release heparin (synthesis it or carry it from mast cells) which has 2 functions:

a) Anticoagulant and fibrinolytic.

b) After fatty meals, heparin secretion increases in the blood to prevent lipemia (it activate lipase enzyme which hydrolysis plasma triglycerides, so prevent and treat atherosclerosis).

2) Secrete histamine to attract the eosinophils (that inactivate histamine).

3- BASOPHILS (HEPARINOCYTE)

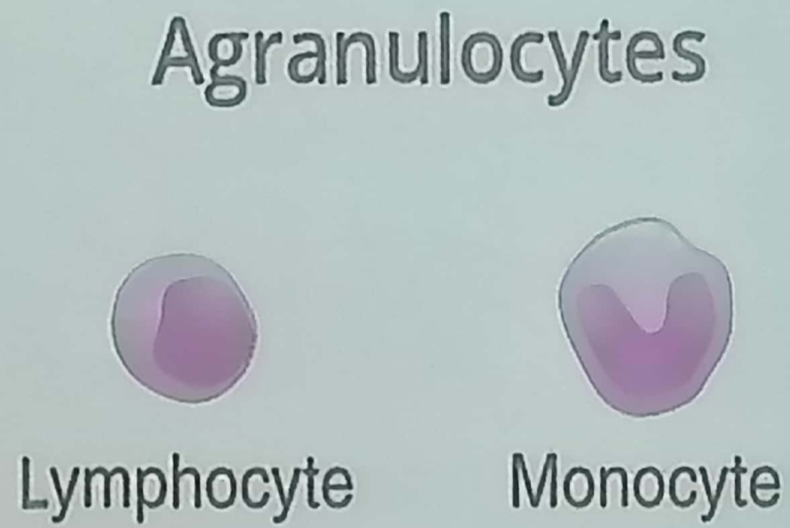
Basophilia: Means an increase in the number of basophils which occurs in chronic inflammation, leukemia, liver cirrhosis and some skin diseases.

B – Agranulocytes

They have no granules in the cytoplasm. Formed in lymphoid tissue (mainly) & bone marrow. Their nucleus is large.

They have 2 types :-

1- Monocyte.
2- Lymphocyte.



1-Monocyte (Macrophage)

- It is the largest leukocyte (12-20μm).
- Their nucleus is kidney or horse shoe shape.
- Its percentage in all species is 5 %.
- Produced in bone marrow, lymph node & C.T., then circulates for 72 h., then enters the tissue and become tissue macrophage (e.g. Kupfer cells of the liver, osteoclasts of the bone, microglia in the brain and dust cell in the lung).
- Their life span in the tissues not known (may be 3 months), they do not re-enter the circulation.

Functions: (2nd line of defense)

1- They are highly phagocytic to large tissue particles as bacteria, protozoa and fungi. They leave blood to the tissue by **diapedesis** to engulf and hydrolysis the large particles.

After the action of neutrophils, the focus of inflammation become acidic (due to accumulation of incomplete oxidized products), which inactivate the neutrophils but the monocyte act in acidic medium and complete the actions of neutrophils

2- Monocyte help tissue repair after inflammation (by it can converted to fibroblast).

Monocytosis:-

It means the increase number of monocytes, which occurs in chronic diseases, leukemia, parasitic diseases (e.g. Malaria) and some infectious diseases (as T.B. and brucellosis).

Pus formation:-

After neutrophils and monocytes engulf much bacteria and necrotic tissue, they die. After several days a cavity is formed in the area of inflammation and this cavity contains pus (which is necrotic tissue, dead neutrophils and dead monocytes).

2- Lymphocytes

- There are 2 forms of lymphocytes, small (5-10 µm) and large form (10-15).
- They have a large spherical nucleus surrounded by a thin cytoplasmic rim.
- Life span about 3-4 days but sometime can be 100-200 days.
- They are formed in lymphoid tissue mainly, some of them are formed in the bone marrow.

Their number varies according to:-

1- Species: 25% in man, dog, cat & horse; 60% in pig, ruminants & birds.

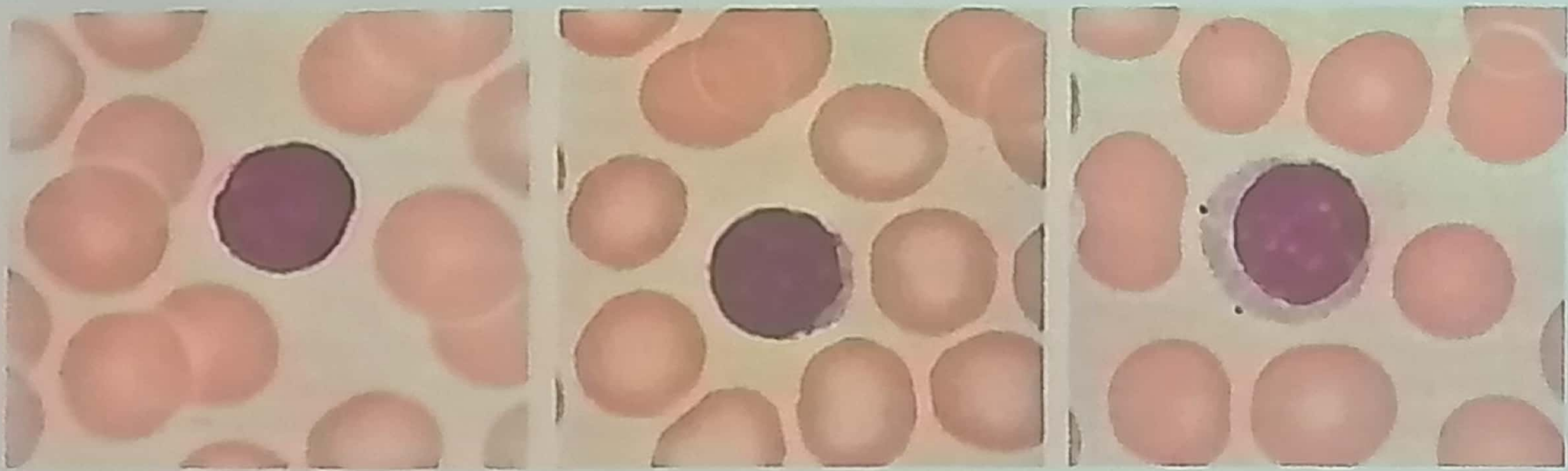
Functions:(3rd line of defense)

1- Antibodies formation.

2-Can change to monocyte (phagocytosis).

3- Can change to fibroblast (tissue repair).

4- ↑ thyroxine.



Lymphocytosis:- It means an increase in the number of lymphocytes, which occurs in neutropenia, after vaccination, hyperthyroidism, T.B., lymphocytic leukemia and ↓ corticoids.

Lymphopenia: It means a decrease in the number of lymphocytes, which occurs in certain viral disease as canine distemper, Aids, stress condition and increase corticoids.

N/L: It is a constant ratio between neutrophils and lymphocytes. It's value about 3 in carnivorous & omnivorous and about 0.5 in herbivorous.

Immune mechanisms

- Lymphocytes are key the constituents of the immune system.
- In mammals, this system has the remarkable ability to produce antibodies against many millions of different foreign agents, that may invade the body.
- In addition, the immune system remembers and a second exposure to a foreign agent produces a more rapid and greater response

Development of the immune system:-

Bone marrow lymphocytes are migrated to:-

Thymus:- → T-lymphocyte : 4 types (memory, killer, inducer (
&suppressor) → **Cellular immunity.**

2) Fetal liver & spleen (or burse of fabricius) → B-lymphocytes (
2 types: memory & plasma cells → **Humeral immunity.**

- Phagocytosis of the foreign body → ↑ killer cells (T_8) → ↑ suppressor cells (T_8) → ↑ inducer cells (T_4) →
  1- ↑ plasma cells to produce γ-globulins (IgM, IgA, IgG, IgE & IgD).
  2- ↑ proliferation of T_8 & plasma cells.

- Acquired immune deficiency syndrome (**Aids**) is a virus disease decreases T_4 cells, leads to loss of T_4, which leads to failure of the proliferation of T_8 & B cells.
- This leads to loss of body immunity and death occur even from normal non-pathogenic bacteria.

Plasma Proteins

- Each 100 ml plasma contains about 7 g proteins.

* **Plasma proteins are** :-

1- **Albumin** : 4 g %,70.000 M.W, synthesis in the liver.

2- **Globulins** :2.7 g %, 150.000 M.W, synthesis in the liver, lymphnodes, bone marrow & spleen. They are many types: α_1,α_2, β_1,β_2 and γ- globulins.

* **A/G ratio**.

3- **Fibrinogen** :0.3 g %, 400.000 M.W, synthesis in the liver.

Functions of plasma proteins:-

1- Blood clotting.

2- Regulation of blood volume.

3- Buffering action.

4- Defense.

5- Maintenance of arterial blood pressure.

6- Maintenance of capillary permeability.

7- Carriers (transport, temporal storage, detoxification and prevent loss from the kidney).

8- Carriage of Co_2.

9- Regulation of erythrocytic sedimentation rate.

10- Reserve proteins.